

Finn Köhler

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EDUCATION

University of California, Berkeley

Berkeley, California, USA

Graduate Student - Computer Science & Engineering (Visiting Student)

Aug. 2026 – Dec. 2026

- Coursework (subject to availability): Vehicle Dynamics & Control (ME 236C), Advanced Robotics (CS 287), Introduction to Embedded Systems (CS C249A), Dynamics and Control of Autonomous Flight (ME 236U).

University of Münster

Münster, Germany

Master of Science in Information Systems (Major in Data Science)

Oct. 2025 – Mar. 2028

- Current GPA: **1,5** (US equivalent $\approx$ **3.7**)
- Coursework: Unsupervised Learning, Optimization and Decision Making, Mining Massive Datasets, Computer Vision, Deep Learning, Supervised Learning
- Focus Areas: Machine Learning for autonomous systems.

Bielefeld University of Applied Sciences (HSBI)

Bielefeld, Germany

Bachelor of Engineering in Digital Technologies (Dual-Study Program)

Aug. 2020 – Feb. 2024

- GPA: **1,8** (US equivalent $\approx$ **3.5**); Thesis: **1,3** (US equivalent $\approx$ **3.8**)
- Thesis: Automated integration of TwinCAT Analytics Dashboard into existing industrial visualizations.
- Selected Coursework: Algorithms & Data Structures, Machine Learning, Cluster Computing, Big Data, Data Mining, Operations Research, Speech & Image Recognition, Quality Assurance for AI Systems.

EXPERIENCE

Application Engineering Intern (Incoming)

Jan. 2027 – May 2027

Beckhoff Automation USA

Silicon Valley, USA

Software Engineer (Working Student)

Oct. 2025 – Aug. 2026

Beckhoff Automation GmbH & Co. KG (TwinCAT Analytics)

Münster, Germany

- Built TwinCAT CoAgent MCP servers for the measurement suite, automating workflows across engineering tools.
- Engineered the TwinCAT Global Watchlist, a real-time PLC monitoring interface supporting **hundreds** of variables at cycle times down to **1 ms**, with snapshot capture/restore and live write-back to running PLCs.
- Developed a Visual Studio service for PLC self-registration, letting users add any variable to the real-time watchlist with a single right-click.

Software Engineer (Full-time)

Feb. 2024 – Oct. 2025

Beckhoff Automation GmbH & Co. KG (TwinCAT Analytics)

Verl, Germany

- Built a generic filtering framework in C# with a custom query language for tree views containing **hundreds of thousands** of items, using reflection and attribute-driven design for plug-and-play extensibility.
- Achieved real-time UI filtering with **millisecond latency** through aggressive caching and constraint-based evaluation.
- Initiated development of the TwinCAT Global Watchlist for real-time PLC diagnostics and visualization.

Dual-Study Software Engineer

Aug. 2020 – Feb. 2024

Beckhoff Automation GmbH & Co. KG (TwinCAT Analytics)

Verl, Germany

- Bachelor Thesis (Grade 1,3 (US equivalent $\approx$ 3.8)): *Automated integration of the TwinCAT Analytics Dashboard into existing industrial visualizations.*
- Architected a synchronization framework unifying legacy visualization concepts with modern TwinCAT Analytics dashboards, and automated User Management configuration for secure server-side data access in TwinCAT HMI.
- Developed Windows kernel-mode drivers in C++ and industrial IPC diagnostics pipelines using the Beckhoff MDP library and asynchronous TwinCAT ADS communication.

RESEARCH

Feed-Forward Multi-Traversal Driving Scene Reconstruction

UC Berkeley (DeepDrive), Mercedes-Benz R&D North America

Jul. 2026 – Present

- Delivered a pilot benchmark comparing **Wild3R** against **AnySplat** on MTGS (Multi-Traversal Gaussian Splatting), establishing baseline reconstruction quality across repeated drives of the same scene.
- Developing a feed-forward 3D Gaussian Splatting pipeline for multi-traversal driving scene reconstruction, incorporating multi-modal sensor priors to improve robustness to transient objects and reconstruction fidelity.

Part-Based Rigid Pedestrian Decomposition in Driving Scene Reconstruction

PyTorch, 3D Gaussian Splatting, SMPL, Waymo Open Dataset

Apr. 2026 – Jun. 2026

- Developed a part-based pedestrian model for the DriveStudio scene-graph framework, decomposing each pedestrian into **10** SMPL-driven rigid body segments with trainable per-segment pose residuals, recovering limb articulation purely through structure, with no per-Gaussian skinning or deformation networks.
- Implemented SMPL forward kinematics, per-segment-aware Gaussian densification, and integrated the new **PartRigidNodes** type across the data loader, trainer registry, and CUDA rasterization pipeline (PyTorch).
- Improved Human-only PSNR by **+3.69 dB** over the rigid-pedestrian baseline (**25.70** → **29.39**) on close-range Waymo scene 23, with visibly coherent leg articulation, at no measurable extra training cost.

SELECTED PROJECTS

Dartz - Multiplayer Darts Scoring Platform | Next.js, ASP.NET Core, PostgreSQL, SignalR 2024 – Present

- Architected a multiplayer scorekeeping platform with a dedicated ASP.NET Core game server using SignalR and WebSockets for real-time state synchronization across remote clients.
- Implemented authoritative game logic for "501" rules on the server-side with live matches supporting remote and local players, 3D dart models, and player performance analytics.
- **Live demo**

Handwritten Math Symbol Recognition + Grad-CAM | Python, TensorFlow, CNN 2023 – 2024

- Trained and evaluated a CNN to classify handwritten mathematical symbols (**80+** classes) using a Kaggle dataset with **100k+** images.
- Achieved **93.6%** test accuracy and analyzed failure modes via confusion matrices and misclassification clustering.
- Implemented Grad-CAM visualizations to explain model decisions and highlight salient regions used for classification.

Robotics club & Automated Pathfinding 2018 – 2020

- Engineered an autonomous robot capable of real-time pathfinding and object targeting using remote control and onboard logic.
- Organized STEM hands-ons for interested students in local schools.

TECHNICAL SKILLS & ADDITIONAL INFORMATION

Programming: C#/.NET, Python (PyTorch, scikit-learn, pandas, NumPy, matplotlib), C/C++, TypeScript (React), SQL

AI & Data: Machine Learning, LLM pipelines, Deep Learning, Data Mining, Hadoop, Spark, MCP Servers, Agents

Automation & Real-Time Systems: TwinCAT PLC (IEC 61131-3), TwinCAT Analytics, TwinCAT HMI, TwinCAT ADS, Industrial IoT pipelines, Windows kernel-mode development, MQTT

Technologies: Git, Docker, Firebase, Hadoop, Spark

Methodologies: Agile/Scrum

Languages: German (Native), English (TOEFL iBT: **117/120**), Spanish (Basic)

Leadership: Handball Team Captain; Elected Member of Computer Science Student Council

Honors & Affiliations:

- **Fulbright Travel Scholarship Nominee:** Recognized for academic excellence and cross-cultural exchange (Declined to attend UC Berkeley).
- **German Informatics Society (GI):** Received the Award for Outstanding Achievement in Computer Science alongside an honorary membership.